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Forest Ecology and Management 208 (2005) 223–236

Forest Ecology and Management

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Changes in structure and composition of maple–beech stands following sugar maple decline in Québec, Canada

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Received 3 August 2004; received in revised form 1 December 2004; accepted 3 December 2004

Abstract

The sugar maple decline observed in forests of northeastern North America is one of the disturbances that may have caused changes in the structure and composition of maple-dominated stands. Such changes may alter the sustainability of these uneven-aged ecosystems. The hypothesis that the decline observed in sugar maple populations over recent decades has modified stand structure and composition was tested using data from the Forest Ecosystem Research and Monitoring Network (RESEF) in Québec, Canada. Analysis of the diameter class density distribution, growth rates, and mortality schedule for 12 stands indicated a decline in health and productivity in the sugar maple population. This result suggests that environmental factors have limited the development of this species in all strata. The American beech population responded by doubling its pole cohort density in 10 years. Although increases in American beech stem density tended to counter sugar maple mortality, results suggest that stand structure has been shifted from a sustainable structure to an unsustainable one over a 10-year period. The high mortality of sugar maple and the large increase of pole-size beech trees deeply altered forest structure and composition in the majority of stands studied. These changes question seriously the sustainability of sugar maple in these forest ecosystems.

Keywords: *Acer saccharum*; *Fagus grandifolia*; Forest dynamics; Diameter distribution; Rotated sigmoid; Forest health

1. Introduction

Sugar maple (SM; *Acer saccharum* Marsh.) has a broad distribution in North America, and thrives throughout the northeastern United States and southeastern regions of Canada. The bioclimatic domain of SM-dominated stands covers approximately 17% of the commercial forest area of Québec, Canada (Robitaille and Saucier, 1998). In this region, this species accounts for roughly 36% of the industrial transformation of hardwood species (Gouvernement du Québec, 2002a). Moreover, the province of Québec is the most important producer of sugar maple syrup, accounting for 70% of the world production (Gouvernement du Québec, 2002b).

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0378-1127/$ – see front matter
doi:10.1016/j.foreco.2004.12.003

Large parts of the SM forest stands in this region are uneven-aged, and secondary species are generally American beech (AB; *Fagus grandifolia* Ehrh.) and yellow birch (YB; *Betula alleghaniensis* Britton), typical of the northeastern hardwood forest (Leak, 1987).

The contemporary composition and structure of the northeastern hardwood forest has been determined by a combination of historical endogenous and exogenous disturbances (Bormann and Likens, 1979; Oliver, 1981), interspecific differences in shade tolerance (Shugart, 1984; Connell and Slatyer, 1977), and site characteristics, notably soil resources (Van Breemen et al., 1997; Fahey et al., 1998). Historical forest management such as widespread clearing and partial cutting in northeastern North America during the 19th and early 20th centuries profoundly altered the landscape structure. Since then, much of this territory has been naturally reforested, resulting in changes in the composition and structure of the forest that defines the present-day forest vegetation (Simard and Bouchard, 1996; Dyer, 2001).

Recently, SM decline and dieback have emerged as important factors structuring hardwood forests. Despite the absence of scientific consensus on the nature of the perturbation, symptoms of SM decline and dieback have been observed at different scales in the deciduous forest of northeastern North America since the late 1970s (Kolb and McCormick, 1993; Horsley et al., 2000), and particularly in Québec (Duchesne et al., 2002, 2003). The poor SM health status may have caused a modification in forest structure. It has been previously suggested that gap formation in the forest canopy, or even a change in light interception as a result of the decline of SM, may increase growth and establishment of tree seedlings (Houle, 1990; Perkins et al., 1987; Pothier, 1996) leading to a reorganization of the forest system.

Using data from the Québec Forest Ecosystems Research and Monitoring Network (Réseau d'Étude et de Surveillance des Écosystèmes Forestiers, RESEF), the main objective of this study was to document the recent changes in composition and structure of unmanaged SM stands in Québec having different rates of decline. We wanted to verify the hypothesis that SM decline has provided an opportunity for another species (particularly AB) to gain dominance in certain size classes, thus altering stand structure sustainability.

2. Materials and methods

2.1. Study sites

The RESEF was implemented in Québec in the mid-1980s, its main goal being to increase our understanding of the response of forest ecosystems to environmental stresses. Plots were established in protected area across the province, such as national parks and wildlife reserves, in order to represent regional, contemporary natural forests, free of signs of major silvicultural disturbances, such as partial- or clearcutting. Twelve maple/beech-dominated stands were selected from the RESEF for the purpose of this study. These stands, monitored for 10 years, are uneven-aged; the secondary species are generally AB and YB. They are located in the southern part of the province and along a line that follows the Saint-Lawrence River Valley (from 45°21′ to 47°44′N latitude and 68°42′–76°06′W longitude). Elevation ranges from 215 to 480 m, slopes from 0 to 15%, and annual precipitation from 840 to 1130 mm. Soils are all podzols or spodosols, with the exception of plots 701 and 702, which are on a distric brunisol or Typic Haplorthod (Canada Soil Survey Committee, 1992; Soil Survey Staff, 1999). Previous studies have shown that these SM-dominated stands experienced various levels of decline (Duchesne et al., 2002). The growth trend of dominant and co-dominant trees has been generally level or declining since the early 1960s, suggesting a continuous growth decline for some stands (Duchesne et al., 2002).

2.2. Sampling

Each study plot (50 m × 100 m, surrounded by a 100 m buffer strip) was monitored every 5 years for 10 years, starting from the 1986 to 1990 plot establishment period. Every living tree reaching a height of 1.3 m above ground level was accounted for and classified according to its species. Trees with DBH > 1.0 cm (diameter at breast height, i.e., 1.3 m above ground level) were tagged, positioned, and their respective DBH was measured using either a diameter tape or micrometric compass to the nearest millimeter.

At the edge of the monitoring plots, foliage of 20 sugar maple trees was collected in the mid-crown on two opposite branches. Foliar sampling was carried out from mid-July to mid-August, when foliar element concentrations are stable (Duchesne et al., 2001). From 10 to 16 samples in the first 30 cm of the mineral B horizon were also taken at the plot boundaries for laboratory analysis.

2.3. Chemical analysis

Leaves were initially dried at 65 °C, and then ground to 250 um. Nitrogen concentration was determined following Kjeldahl digestion (Kjeltec Tecator 1030). For P, K, Ca, Mg and Mn content, 500 mg of foliage were microwave-digested with HNO_3 in Teflon TM bombs. Following digestion, element concentrations were measured by atomic emission spectrophotometry. All soil samples were air dried and sieved to 2 mm. Exchangeable Ca, Mg, K, and Al were extracted with an unbuffered NH_4Cl solution (1N, 12 h), and measured using inductively coupled plasma emission. Hydrogen concentration of the extract was measured using a pH probe. Cation exchange capacity was calculated as the sum of the measured cations.

2.4. Data handling

For each stand, total number of stems and their corresponding basal area were calculated for all species, and for SM and AB separately. Population dynamics among stands was first assessed by examining the relative mortality, the relative change in number of stems, and the basal area over the 10-year period since the plots' establishment. The relative importance of each species on a large-scale perspective was then assessed by summing the relative frequency, density, and dominance of each species according to Kent and Coker (1992).

In terms of structure, Meyer (1952) defined a balanced, uneven-aged forest as one that sustains itself by offsetting current growth with current mortality, while maintaining the structure and volume of the forest. De Liocourt (1898), followed by Meyer and Stevenson (1943) were apparently the first authors to report that diameter distribution in uneven-aged old growth stands were well fitted by a negative exponential distribution, which is characterized by a constant ratio in the number of trees between successive size classes. The balanced concept has been referred to as a sustainable, equilibrium, or steady-state structure. All these terms apparently have a similar meaning: a diameter distribution that will be maintained over time in an unmanaged stand through growth, recruitment and mortality (Leak, 1996). There is also evidence from a number of studies that the stable or equilibrium population structure of undisturbed uneven-aged forests may display a plateau in the mid-diameter range, resulting in a "rotated-sigmoid curve" (West et al., 1981; Goodburn and Lorimer, 1999). The law of Liocourt (balanced concept) was primarily developed for large-scale analysis, and is not generally useful for data from individual stands (Harcombe, 1987).

Density–diameter distribution curves were then constructed for SM, AB, and all species combined, by pooling the 12 plot data based upon the number of trees in all plots divided by the total area. Initial and actual stem diameter distributions were fitted with linear (Liocourt type) and cubic (rotated-sigmoid type) regression models, where the dependant variable was the natural logarithm of trees per hectare and the independent variables were mid-points of the 1.5 cm DBH classes. Distributions were fitted up to the 59.25 cm DBH class, dropping the highest classes because of the highly variable occurrence of these individuals. According to Leak (1996), a cubic regression model denoting the presence of a rotated-sigmoid form rather than a negative exponential form was retained if the model showed a significant coefficient for the DBH midpoint, plus a significant reduction in the sum of squares of errors (SSE) by the addition of the midpoint squared and the midpoint cubed. These conditions shall be coupled with a difference in the sign between the latter two coefficients. Significant reductions in SSE were tested by a procedure outlined in Neter and Wasserman (1974).

The diameter distribution may be used in uneven-aged forest management strategies to determine adequate yield to ensure a sustainable production (Nyland, 1998). As suggested by Manion and Griffin (2001), this distribution may also be used in forest health monitoring. Indeed, the ratio of stem density (N) between two successive DBH classes represents the baseline-surviving fraction of stems passing from class (i) to class (j) (where $j = i + 1.5$) (1):

$$\frac{N_j}{N_i} = \mathrm{SF}_i \tag{1}$$

A baseline-surviving fraction by DBH size class, required to maintain the initial structure, was first assessed based on the first measurement and the selected regression model. The confidence interval of the baseline-surviving fraction was determined by a bootstrap method (10,000 reps) based on the error associated with the selected size distribution model.

The surviving fraction between two successive DBH classes was uniquely determined by the growth rate and the mortality schedule. Knowing both the average annual diameter growth (AGR) for each class (i) and the constant class interval (X) allowed the calculation of the time needed to cross one DBH class. By multiplying the estimated period required to cross a specific DBH size class by the annual mortality rate (AMR) allowed the calculation of a mortality rate for a specific DBH class. Finally, the surviving fraction (SF) was the complement of mortality for a specific size class as calculated by Eq. (2):

$$\mathrm{SF}_i = 1 - \left(\frac{\Delta X}{\mathrm{AGR}_i} \times \mathrm{AMR}_i\right) \tag{2}$$

Observed growth and mortality by DBH size classes were calculated from differences between the final and initial samplings. The observed surviving fraction per size class was calculated from the survey data, and was then compared to the baseline-surviving fraction to maintain initial size distribution (Manion and Griffin, 2001). This method allowed us to assess the influence of growth and mortality rates on the surviving fraction and the resulting changes in forest structure from initial to final distribution.

Foliar norms from the diagnosis and recommendation integrated system (DRIS) (Lozano and Huynh, 1989), soil saturation in exchangeable calcium (Ouimet and Camiré, 1995), basal area increment (BAI) trend of dominant and co-dominant maples (Duchesne et al., 2003), and percent defoliation were previously identified as indicators of site quality for sugar maple. Average BAI trend since the 1960 and the 1990 average percent defoliation for dominant and co-dominant sugar maple trees were assessed by the procedure outlined in Duchesne et al. (2002). Finally, correlation analyses were used to test relationships between indicators of sugar maple health and the change in pole stem density between initial and final sampling. This analysis was carried out because the increase in the density of tree regeneration has been identified as a potential result of sugar maple decline on stand structure (Houle, 1990; Pothier, 1996).

3. Results

3.1. Individual stand dynamics

Initial stand characteristics at the year of plot establishment are presented in Table 1. Basal area ranged from 21.9 to 32.6 $m^2\ ha^{-1}$, for which SM and AB accounted for 40–90 and 2–44%, respectively. Stand density varied from 778 to 2114 stems ha^{-1}, for which SM and AB accounted for 21–89 and 3–66%, respectively. Basal area increment trend since 1960 ranged from -0.33 to 0.06 $cm^2\ year^{-2}\ tree^{-1}$, decline classes ranged from 2.3 to 18.1%, Ca DRIS index from -46.3 to 10.2 and soil saturation in exchangeable calcium from 7.5 to 99.3%.

The relative mortality and relative changes in stand density and basal area for SM and AB 10 years after plot establishment are presented in Fig. 1. The density of SM was reduced in all stands, except for stands 402 and 501, where SM density increased by 3.7 and 12.2%, respectively. On average, SM density decreased by 14.2% over the 10-year period. On the other hand, the density of AB increased in all stands except for stand 101, where its density decreased by 21.2%. Overall, 10 years after plot establishment the density of AB had increased by nearly 1.5 times. The relative mortality varied greatly among stands. The relative mortality of SM over the 10-year period ranged from 1.2 to 3.4%, while the mortality of AB varied from 0 to 1.8%. Overall, the relative mortality was greater for SM (2.0%) compared to AB (0.8%) ($P < 0.001$). Relative changes in basal area were also lower for SM (7.3%) than for AB (16.2%), but variability among stands was too high to detect any statistical differences. The change in basal area varied from -11.0 to 20.0% and from -15.0 to 68.7% for SM and AB, respectively.

Table 1
Selected site characteristics and sugar maple site quality indicators

Site	Year of establishment	Elevation (m)	Humus classification	Stand density[a] (stem ha^{-1})			Basal area[a] (m^2 ha^{-1})			Ten-year change in beech pole density	SM site quality indicators			
				All	SM	AB	All	SM	AB		Average decline class (%)	Basal area growth trend (cm^2 $year^{-2}$ $tree^{-1}$)	Foliar Ca DRIS index	Soil exch. Ca/CEC[b] (%)
101	1988	310	Moder	1098	838	236	29.0	19.6	8.2	−34	3.6	0.06	−2.3	39.0
301	1986	285	Moder	2114	446	1392	28.3	21.5	3.2	1304	18.1	−0.33	−37.6	14.2
401	1987	255	Mor	1888	878	474	21.9	14.9	4.1	64	7.8	n.e	−20.2	49.5
402	1989	275	Mor	1762	810	558	24.8	12.0	10.8	412	2.3	−0.06	−10.0	44.3
501	1988	350	Mor	1638	588	550	32.6	18.6	3.1	654	5.7	0.002	−25.6	54.7
502	1988	480	Mor	1692	728	798	27.3	10.9	7.6	1270	7.4	−0.16	−46.3	7.6
701	1987	335	Mull	788	702	22	27.5	22.5	0.6	44	7.1	n.e	10.2	86.1
702	1988	215	Mull	778	456	32	25.6	16.7	1.1	224	4.7	0.03	0.8	99.3
703	1990	270	Mor	1418	696	168	29.0	20.9	0.8	166	5.7	n.e	−7.5	92.6
1201	1987	530	Mor	912	772	74	25.0	22.5	2.0	94	8.6	−0.06	−19.8	32.2
1202	1987	420	Mor	1210	706	398	30.3	20.1	7.3	534	13.2	−0.02	−44.9	7.5
1501	1987	415	Moder	2032	802	1094	27.0	21.9	4.5	988	7.2	0.03	−16.2	33.3
Average				1444	702	483	26.5	18.0	4.3	477	7.0	−0.037	−17.8	46.7

[a] For trees with DBH $\geq$ 1.1 cm at the year of establishment. All: all species; SM: sugar maple; AB: American beech.
[b] For the first 30 cm of the mineral B horizon.

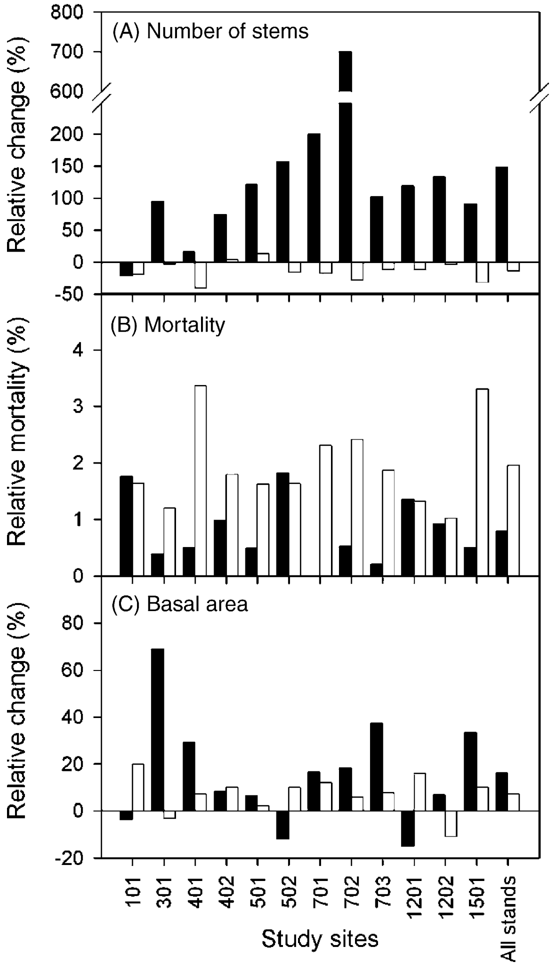

Fig. 1. Relative mortality (B) and relative change in density (A) and basal area (C) for sugar maple (white columns) and American beech (black columns) over 10 years in the selected study plots.

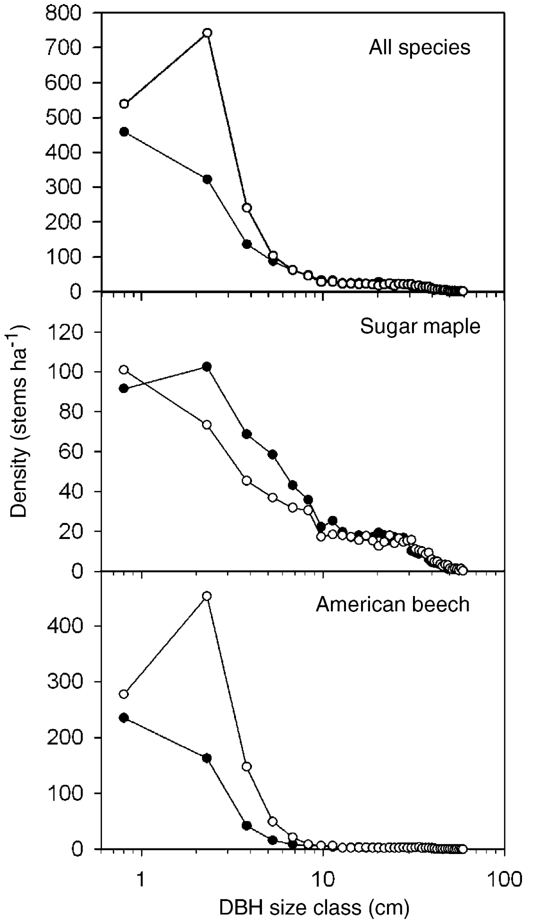

Fig. 2. Initial (black dots) and final (white dots) stand structure (after 10 years) for all species combined, sugar maple, and American beech.

3.2. Large-scale structure analysis

Initial and final density–diameter distribution curves for SM, AB, and all species combined are shown in Fig. 2. In the initial distribution, the number of trees by DBH classes for all species combined was declining throughout its range. However, SM and AB showed very distinct patterns after 10 years, with density of SM stems smaller than 29.25 cm DBH dropping by 17%. At the opposite extreme, the AB pole density (0.75–5.25 cm DBH) was nearly twice the initial distribution. Given that the density of trees in the next diameter class was clearly higher than in the previous class, the final distribution can be qualified as unbalanced (Leak, 1996).

3.3. Large-scale species dynamics

The initial density–diameter distribution curves, the selected model, and the corresponding baseline-surviving fraction for SM, AB and for all species combined are shown in Fig. 3. Diameter distributions were all classified as rotated-sigmoid distributions (Table 2). The baseline-surviving fraction thus showed a parabolic shape. This shape indicated that the surviving fraction of trees passing from one DBH class to another would vary among size classes in order to maintain the initial forest structure. Annual DBH

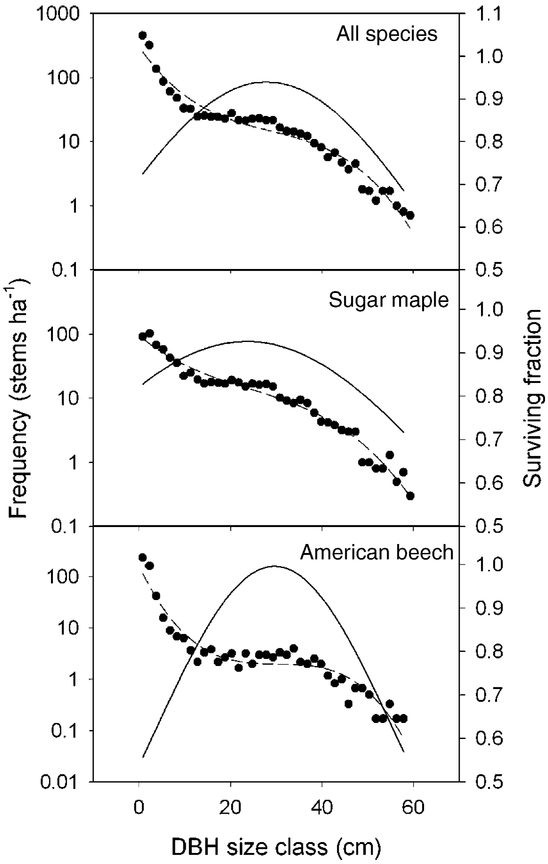

Fig. 3. Size distribution of living trees in the pooled plots (black dots), the best fitted linear model (dash lines) and the associated surviving fraction to maintain structure (solid lines) for all tree species, sugar maple, and American beech.

growth rate by size class for all species and for SM and AB are presented in Fig. 4. All diameter growth trends were parabolic in shape. The growth rate for SM was lower than for AB in the lower DBH classes, while they were more similar in the mid-size classes. The growth rates of the first four DBH classes of AB showed higher values than suggested by the parabolic curve. This resulted in a breakpoint in small classes of the diameter distribution when all species were combined. On average, annual DBH growth rates were 0.15, 0.14 and 0.15 cm year^{-1} for all species, SM, and AB, respectively (Table 3).

Annual mortality rates by DBH class showed a parabolic trend for all species and for SM only, for which high mortality rates occurred in both the large- and small-diameter trees (Fig. 4). American beech showed a distinct pattern for which the mortality rates were nearly levelled for the 0.75–30.75 cm DBH classes, and increased subsequently in the higher classes. Overall, average mortality rates were 1.9, 2.5 and 1.0% for all species, SM, and AB, respectively.

The observed surviving fraction and its theoretical baseline needed to maintain initial structure are shown in Fig. 5. Comparative analysis of SM and AB showed that these species exhibit distinct patterns. The surviving fraction of SM in the lower DBH classes (0.75–14.25 cm) was clearly below the baseline-surviving fraction. Opposite to this, the AB pole (0.75–9.75 cm DBH) surviving fraction was superior to the baseline values. When all species were considered, the increase in the AB pole density partially offset SM mortality in the first three DBH classes. However, for the 5.25–14.25 cm DBH class, the secondary species density was not sufficient to offset SM mortality. For those size classes, we observed a lower surviving fraction than the baseline values needed to maintain the initial structure.

The higher mortality and lower growth rate of SM trees as compared to AB modified the relative

Table 2
Adjusted R^2 and mean square error (MSE) of estimates for linear and polynomial regressions on logarithmic number of stems by hectare over DBH class at the year of establishment (initial) and 10 years later (final)

Species	Initial model					Final model				
	Linear		Cubic		SSE reduction test	Linear		Cubic		SSE reduction test
	R^2	MSE	R^2	MSE		R^2	MSE	R^2	MSE	
All species	0.91	0.23	0.95	0.11	$F = 20.3$; $P < 0.001$	0.85	0.38	0.94	0.15	$F = 30.6$; $P < 0.001$
Sugar maple	0.92	0.17	0.96	0.08	$F = 22.8$; $P < 0.001$	0.85	0.30	0.95	0.10	$F = 39.0$; $P < 0.001$
American beech	0.76	0.63	0.90	0.26	$F = 28.3$; $P < 0.001$	0.73	0.87	0.91	0.29	$F = 39.3$; $P < 0.001$

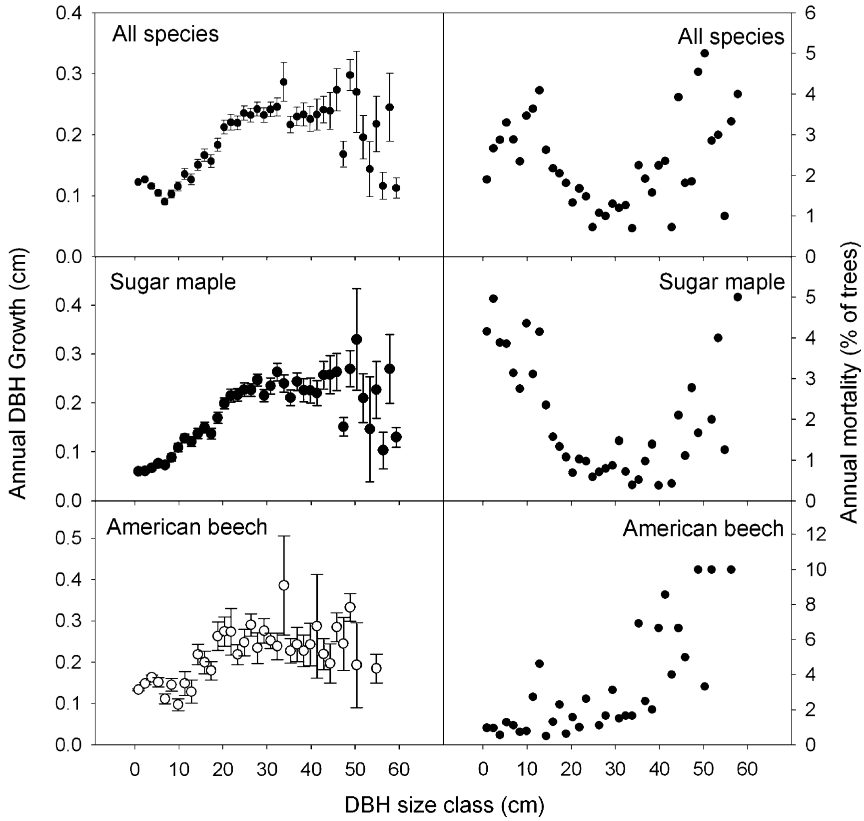

Fig. 4. Annual growth and mortality rates by DBH class for all species, sugar maple and American beech. Error bars = ±standard error.

importance of those species (Table 3). Relative importance of SM decreased by 15.7% while AB increased by 23% in 10 years.

3.4. Changes in beech pole density versus sugar maple health indicators

Change in beech pole (DBH < 9.1 cm) density between initial and final sampling, and indicators of site quality for sugar maple are shown in Table 1. The change in beech pole density was negatively related to the BAI growth trend since 1960 ($r = -0.69$, $P = 0.039$), foliar Ca DRIS index ($r = -0.71$, $P = 0.009$), and soil saturation in exchangeable calcium ($r = -0.61$, $P = 0.037$). Changes in beech pole density also showed a marginally positive correlation with the defoliation class ($r = 0.52$, $P = 0.084$).

4. Discussion

4.1. Structural changes

The rotated-sigmoid distribution was formerly considered as the natural steady-state distribution in unmanaged stands (Goff and West, 1975; Goodburn and Lorimer, 1999) or, by others, as the result of earlier disturbances (Schmelz and Lindsey, 1965; Leak, 1996, 2002). In the present study, initial diameter distribution was classified as a rotated sigmoid (Table 2). Sustainable diameter distributions have one unique characteristic: the curve created by the number of trees over DBH is declining, or occasionally level, throughout its range (Leak, 1996). According to this definition, diameter distribution for all species was sustainable in our stands in the beginning. Based on DBH of the older trees included

Table 3
Number of sampled trees, relative importance, and DBH growth rate for each species

Species	Number of samples	Relative importance		DBH growth rate ± S.E. (cm $year^{-1}$)
		Initial	Final	
All species	7108			0.15 ± 0.001
A. saccharum Marsh.	3213	42.6	35.9	0.14 ± 0.002
F. grandifolia Ehrh.	2686	20.4	25.1	0.15 ± 0.002
B. alleghaniensis Britton	301	6.7	6.7	0.17 ± 0.008
Acer pensylvanicum L.	267	4.7	6.3	0.12 ± 0.004
Ostrya virginiana (Mill.) K. Koch	149	2.3	3.4	0.11 ± 0.005
Acer rubrum L.	125	4.1	4.1	0.19 ± 0.014
Tilia americana L.	100	2.6	2.6	0.13 ± 0.009
Tsuga canadensis (L.) Carr.	67	1.7	1.7	0.19 ± 0.013
Abies balsamea (L.) Mill.	58	2.6	2.6	0.17 ± 0.011
Picea rubens Sarg.	49	2.2	2.1	0.13 ± 0.012
Fraxinus americana L.	46	1.9	1.8	0.27 ± 0.027
Other species[a]	47	8.2	7.7	0.18 ± 0.013

[a] Other species included *Betula papyrifera* Marsh., *Prunus pensylvanica* L.f., *Quercus rubra* L., *Acer spicatum* Lam., *Fraxinus nigra* Marsh., *Juglans cinerea* L., *Ulmus Americana* L., *Populus tremuloides* Michx., *Prunus pensylvanica* L.f., *Picea glauca* (Moench) Voss, *Pinus strobus* L.

in the distribution, this sustainable structure may have been maintained for roughly 400 years (60 cm at 0.15 cm $year^{-1}$ = 400 years). Establishment of an AB pole cohort resulted in an unbalanced distribution, where the density of trees in the next diameter class was greater than in the previous class (Fig. 2). If the AB cohort continues to develop, the distribution will result in a near bell-shaped distribution. A bell shape cannot be sustained, either naturally or through a cutting regime (Leak, 1996). Therefore, these trends suggest that stand structure has been temporarily shifted from a sustainable structure to an unsustainable one over the 10-year period.

The sigmoid shape was more pronounced for AB than for SM, as expressed by the *F* values associated to the SSE reduction test (Table 2). It was also more pronounced for the final than for the initial distribution. This evolution was mainly the result of the increased AB pole density following growth and mortality rates of each species. According to Leak (1996), the development of a more rotated-sigmoid distribution suggests a disturbance event favoured the development of a rotated-sigmoid distribution.

4.2. Species dynamics

For SM, a slower growth rate combined with a higher mortality rate resulted in a lower surviving fraction than the baseline values needed to maintain the initial structure in the 0.75–14.25 cm DBH size classes (Fig. 5). The situation, where the observed final surviving fraction was lower than the baseline, suggests health and productivity problems for this species (Manion and Griffin, 2001). At the other extreme, the higher growth rate of saplings combined with the lower mortality rate for AB resulted in a final surviving fraction higher than the baseline value for the first seven DBH classes. This result indicates either a changing forest structure, or an instability leading to future mortality above the baseline level required to bring the forest structure back into balance (Manion and Griffin, 2001).

A sustainable forest ecosystem is one that maintains a stable size structure relationship by balancing growth with mortality (Manion and Griffin, 2001). With respect to resource limitations, the increase in AB pole density can only be explained by the reduced resource requirement of SM, for which DBH was larger than the AB cohort. Reduced resource requirements may be partially attributable to the slower growth rate and higher SM mortality in the lower DBH classes (0.75–14.25 cm). For higher SM classes (>14.25 cm), a previous study showed that dominant and co-dominant SM growth trends, indicators of tree vigour, have been declining for the majority of these stands since the early 1960s (Duchesne et al., 2002; Fig. 6). It appears that these older trees were in senescence despite the absence of departure between the observed and the theoretical surviving fractions to maintain initial stand structure.

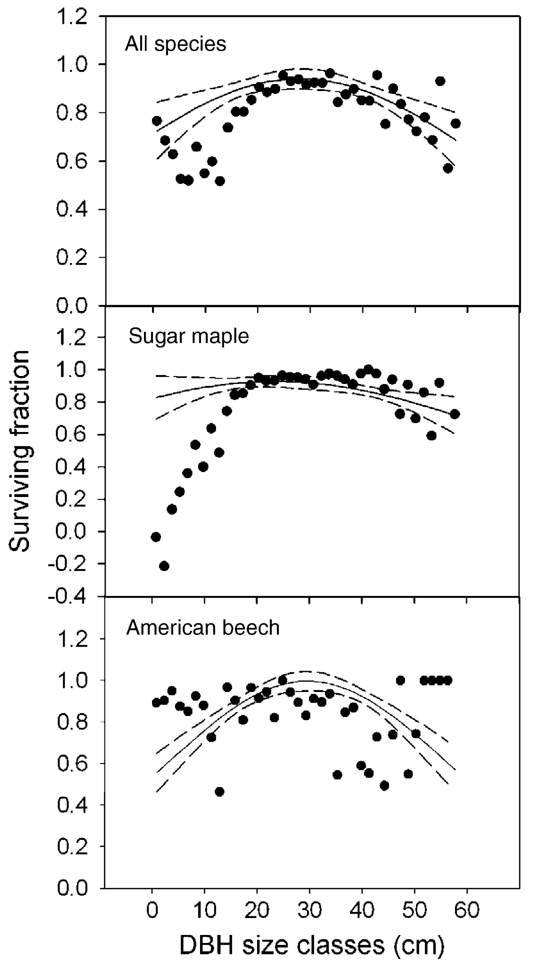

Fig. 5. Observed surviving fraction 10 years after initial measurements (black dots), calculated baseline-surviving fraction derived from these initial measurements (straight lines), and their 99% confidence intervals (dashed lines) for all species, sugar maple, and American beech.

The longer duration of the senescence period for older trees as compared to young individuals may explain the stable mortality rate and, consequently, the stability in the surviving fraction for these individuals.

4.3. Changes in beech pole density versus sugar maple health indicators

The increase in AB pole density is attributed to the lower vigour and higher mortality rate of SM trees. The slower growth, higher mortality, and resulting

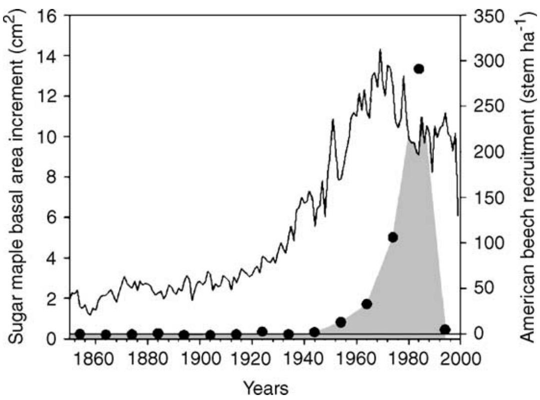

Fig. 6. Basal area growth trend of dominant and co-dominants sugar maple trees (straight line) and estimated timing of recruitment of AB individuals (shade area) deduced from the discrepancy between initial and final distribution and average AB diameter growth rate.

surviving fraction of SM suggest that the environmental conditions were no longer adequate for SM recruitment and development. The sudden increase in AB pole density over the 10-year period was negatively related to selected SM site quality indicators. These results suggest that, despite any distinct departure in the observed surviving fraction, the reduced consumption of resources (i.e., light, nutrients, water) by SM in the higher DBH size classes contributed to the development of the AB pole cohort. The increase in AB pole density was limited to the lowest DBH classes (0.75–5.25 cm). From average DBH growth, timing of recruitment of AB individuals contributing to the discrepancy between initial and final distribution may be estimated (Fig. 6). We deduced that AB recruitment was initiated around 1960 (5.25 cm = 35 years at 0.15 cm year^{-1}) and peaked around 1980 (Fig. 6). The AB pole cohort establishment is synchronized with the observed decreased trend in growth of SM trees initiated in the early 1960s (Duchesne et al., 2002, 2003) while the early 1980s peak is associated with high defoliation rates triggered by an extreme climatic event in the winter of 1981 (winter thaw followed by deep freezing) (Payette et al., 1996).

The change in beech pole density was also negatively related to indicators of SM nutrient status (i.e., DRIS index and soil saturation in exchangeable calcium). There is evidence that the distribution of SM is related to the availability of soil mineral nutrients

(Whitney, 1991; Van Breemen et al., 1997). Sugar maple develops best in deep, fertile, moist and well-drained soils (Horsley et al., 2002). Many studies have shown that, for non-calcareous soils, acid deposition has increased leaching losses of basic cations above the replenishment rate from soil mineral weathering and atmospheric depositions, causing a reduction in the availability of mineral nutrients (Houle et al., 1997; Ouimet et al., 2001). Poor soil quality due to low soil nutrient availability and aluminium toxicity were deemed as among the causal factors of SM decline in Québec (Ouimet and Camiré, 1995; Moore et al., 2000; Duchesne et al., 2002). Other authors have observed a significant increase in the survivorship of juvenile SM on calcareous soils, in comparison to nutrient-poor and acid soils in northwestern Connecticut and central-western Michigan (Kobe et al., 1995). It has been demonstrated that soil exchangeable Ca depletion and its influence on seedling dynamics could lead to substantial decreases in SM canopy dominance within a single forest generation (Kobe et al., 2002). Soil acidification also increases Mn mobilisation. High Mn concentration may be toxic or may even limit Ca and Mg uptake by SM seedlings (McQuattie and Schier, 2000). At the opposite extreme, the relative dominance of AB has been inversely related to soil pH and calcium availability (Arii and Lechowicz, 2002). Consequently, AB appears to be less sensitive than SM to soil calcium availability and aluminium toxicity. Although some relationships exist between these variables, cause–effect relationships cannot be tested in such an exploratory study. However, observations suggest that soil properties may no longer be appropriate for establishment and maintenance of SM, and they fostered the growth of a more adapted tree species, American beech.

Sugar maple nutrition is relatively demanding in terms of Ca. For this species, a forest floor mainly formed from SM litter fall contains more than 60% of Ca in terms of exchangeable cation concentrations (Ouimet and Camiré, 1995). At the opposite extreme, the forest floor or even leaf litter under AB foliage contains very low amounts of Ca, while Na is the predominant exchangeable cation (Coté and Fyles, 1993; Dijkstra et al., 2001). Following AB development, the forest floor will become poorer in terms of Ca availability. This may further contribute to SM sapling mortality and regeneration failure.

4.4. Other evidence

Other studies that focused on silvicultural practices have also documented an increase in the number of AB poles since the mid-1980s in both controls and thinned SM stands (Pothier, 1996; Majcen, 1995, 1996, 1998, 1999). Pothier (1996) found an increase of about three-fold in AB pole density in control plots; this phenomenon was similar to the response of the pole cohort 10 years after removing 22 and 35% of stand basal area. This structural change in control plots was attributed to the mortality in the higher size classes and the increased light penetration following the SM decline episode. Indeed, the dieback of branches of large SM trees contributes to a reduction of light interception, and more radiation becomes available for the lower strata. It has been demonstrated that photosynthetically active radiation below the canopy was significantly higher in declining maple stands than in healthy ones, consequently affecting understorey development (Houle, 1990; Perkins et al., 1987).

4.5. Late successional dynamics and other suggested causes

Many studies have noted an increase in the abundance of SM through time, at the expense of AB, when compared to the forest composition in pre-settlement times (Simard and Bouchard, 1996). Increases in SM since then was thought to have been caused by harvesting-related disturbances, suggesting that these relatively young forest stands had not yet reached the late successional equilibrium (Siccama, 1971). Many ecologists have suggested that AB is coming back at the expense of SM in the late successional stage in the northeastern US (Forcier, 1975) and Québec (Beaudet et al., 1999). Some mechanisms have been proposed to explain this shift in forest composition, notably: (1) a better adaptation of AB to grow and survive in the deep shade of old-growth stands (Canham, 1988); (2) the ability of AB to reproduce itself vegetatively by root suckering, thus favouring its regeneration in a shady underscore (Jones and Raynal, 1988); (3) a change in soil properties resulting from an increased beech density in forest stands, which may be detrimental to SM regeneration (Dijkstra et al., 2001); (4) phytotoxicity of AB leaf leachate to SM regeneration (Hane et al., 2003); (5) herbivore preference for SM in comparison to AB (Marquis and Brennemann, 1981).

All these mechanisms may influence regeneration, and consequently, the composition of sugar maple stands over the long term. However, these mechanisms are not suspected of altering the sustainability of forest structure. Therefore, in the present study, the increase in AB pole density and the resulting change in stand structure are attributed to the lower vigour and higher mortality of SM trees in all strata. The results suggest that some environmental conditions are no longer adequate for SM recruitment and development. Thus, the mechanisms involved in late succession of AB may only play a role in regeneration dynamics, but cannot explain the low vigour and high mortality of SM in the higher DBH classes. Consequently, other environmental factors, rather than interspecific competition by secondary species like AB, seemed to have suppressed the development of the SM in all strata, causing massive recruitment of AB and displaced structure from its original dynamics to a disequilibrium state.

Similar increases in density of understorey trees have been previously associated with increased light penetration following an infestation of beech bark disease (Hane, 2003). However, in our study, the beech bark disease has been identified in only 4 of the 12 stands. Moreover, in two of these sites, only one individual was reported to show symptoms of declining health that were attributed to the disease.

5. Conclusion

Following the assessment of growth, mortality rates, and the resulting surviving fraction across diameter classes, our results suggest that SM decline and dieback contributed to bring about changes in SM stand structure and composition. It appears that SM sustainability on relatively poor sites has become a concern in some of its range in Québec. Other secondary species, notably AB, responded to SM mortality by increasing their sapling density. This study also suggests that the recent augmentation of AB poles is not a consequence of its ability to perform under the deep shade of old growth, but rather is a result of fast colonization by an opportunistic species that is better adapted to site conditions following SM decline and dieback. The relatively rapid changes observed suggest that forest dynamics and composition responded to recent changes in environmental conditions, rather than by historical disturbances and interspecific interactions. Our observations also suggest that anthropogenic inputs of sulfates and nitrates and subsequent depletion of base cations and acidification of soils, are the most likely environmental disturbance resulting in the observed destabilization of the SM stands in question. In this context, further studies are needed to explore forest management strategies and possible consequences on biodiversity in these ecosystems.

Acknowledgments

This research was supported by the Ministère des ressources naturelles, de la Faune et des Parcs (project no. 0200 3068). We would like to thank B. Toussaint and J. Martineau for field assistance, and anonymous readers for their helpful comments on an earlier version of the manuscript.

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